

EMI Challenge^[1]

Goal: Predict six continuous emotional mimicry intensity dimensions from in-the-wild multimodal video clips.

Target emotions:



Key challenges:

- Long-tailed sequence lengths
- Sparse and imbalanced labels
- Missing or incomplete text modality
- Noisy in-the-wild audio-visual signals

Sys.	Mods.	Fusion	Split	Val.	Test
F1	T+A+V	MLP	2:1	0.4373	0.506
F2	T+A+V+M	MLP	2:1	0.4386	0.506
F3	T+A+V	MLP	4:1	0.4717	0.570
F4	T+A+V+M	MLP	4:1	0.4722	0.562
F5	T+A+V	MLP (frz.)	2:1	0.4400	0.491

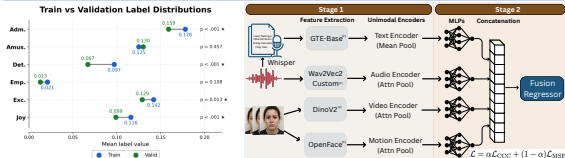
Our submission: Fusion performance across modality combinations and data splits.

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Methods and Materials



Dataset split shows sparse label distributions.

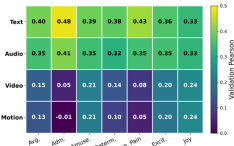
Our two-stage framework: unimodal feature learning followed by multimodal fusion.

Results

Modality	Input / Feature	Model	Val. Pearson
Visual	VIT face embed.	GRU	0.1084
	DINOv2 full-frame embed.	GRU	0.1343
	DINOv2 full-frame embed.	LSTM	0.1260
	VIT face embed.	LSTM	0.1134
	DINOv2 face embed.	GRU	0.1512
	DINOv2 face embed.	Attention pooling	0.1535
Audio	wav2vec2 embed.	Linear projection	0.1428
	wav2vec2 embed.	LSTM	0.3088
	Custom wav2vec2 embed.	Mean pooling + MLP	0.3500
	Custom wav2vec2 embed.	Mean+std pooling + MLP	0.3600
	Custom wav2vec2 embed.	Attention pooling	0.3600
	Custom wav2vec2 embed.	3-layer BiLSTM + attention	0.3650
Text	GTE embed.	LSTM	0.4070
	GTE embed.	MLP	0.4076
	BERT-Large embed. (CLS)	MLP	0.3400

Validation performance of individual modality encoders.

Unimodal Performance Across Emotion Dimensions



Modalities provide complementary cues while also reinforcing common affective patterns.

Discussion

- Text and audio** were the **strongest** unimodal predictors, showing the importance of semantic and vocal cues.
- Visual and motion features were weaker alone but still provided **complementary** affective information.
- The **4:1 split improved performance**, suggesting fusion benefits from more supervised training data.
- Motion gave only modest gains, indicating that **temporal facial dynamics need better modeling**.
- First and second placed teams used stronger fusion/training strategies such as:
 - text-anchored cross-attention,
 - missing-modality tokens,
 - auxiliary supervision
 - VAD-aware audio modeling.

Conclusions

- We proposed a two-stage multimodal framework for EMI prediction using text, audio, vision, and motion.
- Multimodal fusion improved performance over unimodal models.
- Our best system achieved **0.57 test Pearson** and ranked **third** in the EMI challenge.
- Future work can explore stronger motion representations and more adaptive fusion methods.

References

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